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Development of a Through-Tubing Smart Completion Solution

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Abstract

This paper presents the development and qualification of a novel wired and powered tubing/casing system designed to enhance smart completions. It outlines key use cases, the engineering workflow, and extensive component validation that led to achieving Technology Readiness Level (TRL) 4, as per the API TRL scale. It also describes the ongoing planning and preparation for workshop testing to advance to TRL 5 and subsequent field trials for TRL 6.

The development followed the API TRL framework. System requirements were defined through market research and use case analysis, targeting improvements in reliability and cost over existing smart completions. Hardware was designed using CAD, FEM, and CFD tools and fabricated in-house. Multiple iterations of pipe ID evaluation, including MBG testing, led to a refined connection design that maintains mechanical integrity while accommodating powerline for production kits. Prototype tubing strings were equipped with the new components and subjected to analytical simulations and bench tests.

The new system introduces a protected cable pathway inside the tubing using an internal galvanic slip-ring type connector and conduit. This design eliminates the need for external cable connectors and protectors, reducing installation time and minimizing non-productive time (NPT) during completions. Traditional smart completions often suffer from cable damage or installation delays due to external cable routing. This solution addresses those challenges by relocating and shielding the cable internally.

To validate the concept, various software simulations were conducted. While simulations and component tests confirm the design's theoretical feasibility (TRL 4), upcoming workshop testing (part of the future work) will validate real-world assembly processes prior to advancing to TRL 5. This phased approach ensures risks are mitigated before field deployment.

This work introduces a next-generation smart completion system that improves reliability, reduces NPT, and lowers cost by internalizing and protecting critical cabling. The technology's qualification to TRL 4 demonstrates its feasibility and readiness for further workshop and field validation, contributing novel solutions to the evolving landscape of intelligent well completions.

Introduction

Background on the need for real-time data from downhole sensors

Although many wells are still being monitored using periodic well-test measurements, well-bore gradient surveys, and on-site visits, these methods do not allow continuous control over the well. The lack of real-time data from the well can limit proactive decision making and lead to inefficient operations.

In contrast, the installation of downhole gauges can lead to reduced OPEX, improved HSE, and properly monitored well integrity.

Real-time continuous monitoring of downhole data has become essential when dealing with HPHT wells, deep-water subsea wells, unconventional wells with dynamic and transient flow conditions, or smart wells equipped with downhole inflow control devices (ICDs) or inflow control valves (ICVs) (Gonzales et al., 2015). These real-time data are used for reservoir monitoring and analysis, flow assurance enhancement, production optimization, and other purposes (Webb et al., 2008).

Within Petrobras, a digital twin model has been developed to support well integrity management. This model utilizes real-time data from Christmas Tree systems and the data from downhole pressure/temperature (P/T) sensors to enhance monitoring and decision-making (Aranha et al., 2025).

Alternatively, to using downhole permanent gauges, Centrica Energy utilized a distributed temperature sensor (DTS) system to detect leaks at gas-lift mandrels and packers, while pressure testing the tubing string during well commissioning. The leak detection is based on the Joule-Thompson cooling effect (a temperature drops at leak points). This resulted in saving rig time due to efficient leak point detection (Wood et al., 2010).

Background on the need for Surface-controlled downhole devices

Many downhole devices, when controlled from the surface, lead to production optimization, increased operational efficiency, reduced workovers, and a reduced environmental impact. Gas lift valves are among the downhole devices that benefit from surface control. Traditional injection pressure operating (IPO) valves have limitations, as changes in reservoir conditions and water and gas production can affect the design's efficiency. Injection through a non-production gas lift valve or multi-point injection can significantly decrease system efficiency and oil flow rate.

According to Shaw, 2024 integrating surface-controlled gas lift valves leads to continuous production optimization, greater resilience to sand production, and the ability to alternate between tubing and annulus injection without intervention.

Inflow Control Valves (ICVs), when controlled from the surface, can significantly enhance production efficiency and reservoir management by enabling adjustable choke positions. Power transmission to operate these valves can be achieved through fully hydraulic, electro-hydraulic, fully electric, or even wireless methods. Each method, however, comes with its complexities and limitations.

Current limitations

Real-time data recording and transmission. Despite modern advancements in both sensor data recording and transmission, as well as surface control over downhole devices, limitations persist.

When considering production well equipped with electric submersible pumps (ESP), there is often an absence of discharge pressure sensors. This is often due to the limited clearance between the motors outside diameter and the production casings inside diameter, which is required to fit the hydraulic line. ESP models without pump discharge pressure (PDP) data exhibit a high level of uncertainty, particularly in challenging applications such as gassy wells, wells with a high concentration of solids, or wells with unsteady flow (Narvaez et al., 2023).

Fiber optics systems (e.g., DTS), while effective for distributed sensing, are limited by their sensitivity to motion along the fiber axis and require heavy data processing. Additionally, DTS systems are constrained by

slow update rates and susceptibility to optical losses, which limits their effectiveness in real-time monitoring (Gonzales et al., 2015).

Wireless systems also face notable constraints. Battery life typically ranges from 1 to 6 years and is significantly reduced in high-temperature wells. Additionally, these systems offer only low to moderate data transmission capacity, making them unsuitable for applications that require real-time, high-frequency data (Bouldin et al., 2021).

Surface control over downhole devices. ICVs are an emerging technology in intelligent completions. However, since the tubing-encapsulated cable (TEC) is placed on the external wall of the tubing, it decreases the running-in-hole (RIH) speed and presents a significant risk of damage during the installation procedure (Amin et al., 2025).

The traditional setting mechanism of downhole packers for production wells includes both mechanical and hydraulic setting. The mechanical setting has its limitations, particularly in terms of the necessary tubing string weight and pipe manipulation. In contrast, a hydraulic setting requires dropping a ball into the landing nipple, followed by additional workover time to pump the fluid, build necessary pressure, and activate the packer. The electro-hydraulic setting mechanism also has its limitations, including a complex structure, low reliability, and a lack of large-diameter liquid passage (Yang et al., 2024). Existing electrically set packers typically require an additional wireline/e-line trip, resulting in increased RIH time during workovers.

Powerline for production as a Solution

Powerline for Production (PLP) technology offers a promising solution to current challenges within the industry. This through-tubing solution delivers up to 300 W of electrical power to downhole devices and ensures high-rate data transmission of up to 200,000 bits/sec from the downhole sensors to the surface.

Unlike traditional systems, PLP utilizes pre-installed electric conduits inside each tubing joint. This novel solution increases equipment RIH speed and reduces the risk of damage compared to solutions containing external cables.

This paper presents the current state of PLP technology, benchmarks it against existing solutions, and outlines implementation strategies and future development needs.

Current industry challenges and Objectives of PLP development

Current industry challenges

Before stating the PLP objectives and engineering roadmap, it is essential to outline the current industry challenges it aims to address.

- Artificial lift challenges: Gas lift performance analysis is typically based solely on two-pen recorder chart data, limiting diagnostic capability. The absence of real-time P/T profiles in gas lift wells can delay the detection of downhole issues such as multi-point injection and tubing integrity failures. Ting et al., 2022 highlighted the importance of a real-time temperature profile from DTS in identifying the tubing leak point and multi-point injection, which led to a loss of 300 BOPD in offshore Malaysia.
- Flow assurance issues, such as scale and wax depositions, liquid loading, or corrosion within the production/injection wells, can lead to reduced production, decreased mean time between failures (MTBF) of downhole equipment, and increased operating costs. Real-time P/T profiles inside the well will help the operator detect these issues in a timely manner and secure a prompt reaction to the problem.
- Real-time P/T profiles inside the well also support early detection of well integrity issues, which is critical for CO₂ injection, oil and gas, and geothermal wells. Undetected leaks can pose significant risks to both safety and the environment.

- Smart wells equipped with multiple ICVs, and permanent downhole gauges require multiple control lines. Each control line splice necessitates pressure testing and adds 6–8 hours of rig time per splice. This increases completion complexity, run-in-hole (RIH) time, and demands additional wellhead space (Qureshi et al., 2021).

In summary, current challenges span artificial lift monitoring, flow assurance, well integrity, and complexity. The PLP system is being designed to tackle these issues directly.

The powerline for production (PLP) system introduces a protected cable pathway within the tubing, utilizing an internal galvanic slip-ring type connector and conduit. This design eliminates the need for external cable connectors and protectors, enhancing robustness and deployment efficiency, compared to other solutions on the market.

Objectives of PLP development

The PLP system aims to achieve the following:

- Enable high-bandwidth data transmission (>200,000 bits/s) from various downhole sensors to the surface, enabling predictive maintenance, improved decision-making, and enhanced well control.
- Provide a reliable surface to downhole power delivery (up to 300W) to operate intelligent downhole components.
- Simplify well architecture by replacing multiple external control lines with a single through-tubing conduit.
- Enhance operational efficiency by enabling faster, safer, and more efficient RIH/POOH operations during workover.

Description of equipment and processes

Mechanical design

The powerline for production (PLP) system is composed of several key components integrated within the tubing joint. The key components of the PLP system are illustrated in Fig. 1.

- Box and Pin Bushings – installed within the tubing connection ends (BOX and PIN sides), when connected, they provide a mechanical and electrical connection between the two consecutive tubing joints.
- Mechanical locks – two per joint, forming metal-to-metal seals with the pipe body, fixing, and protecting the conduits.
- Power/Data Conduits – dual conduits (for redundancy), enabling bidirectional power and data transfer between joints.

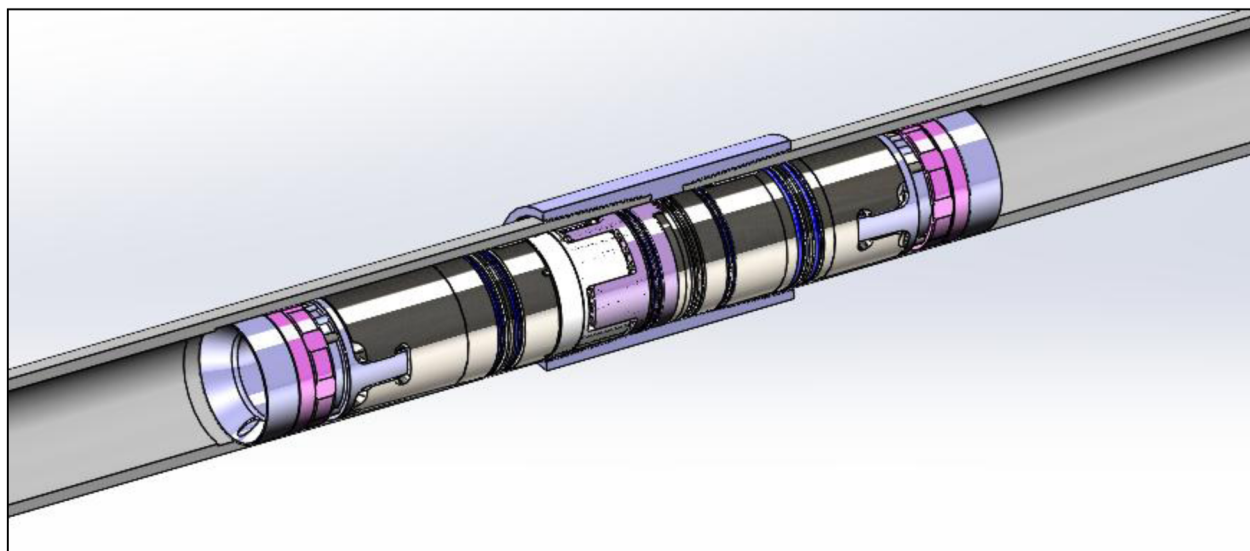


Figure 1—3D model of the powerline for production (PLP) system

Connection process

During the make-up process of the two pipes, the rotating pin bushing aligns and locks with the box bushing in the lower pipe, forming a sealed mechanical and electrical junction. This process is repeated joint-by-joint throughout the tubing string, creating a continuous communication and power path within the tubing string.

Tubing preparation and powerline for production kits installation process

Regarding the high requirements for PLP systems like tight machining tolerances, the chosen connection required to be in the premium performance segment. Therefore, the ISO13679 CAL IV tested and multiple times field proven premium pipe connection was selected.

The performance properties of this connection fit within the required framework of downhole conditions. For PLP usage design adaptations had to be developed, but without reducing the properties of the initial connection. These modifications result in a tailor-made alignment between the dimensions of the produced pipe and the final connection design.

Furthermore, the first MBG testing showed the high influence of minor deformation on the inside diameter caused by interferences between pin and box in the make-up process. The defined interference is indispensable for the performance of the connection, especially for the gas tightness. This leads to incompatibility between the final connection geometry and the high-pressure sealing. This information was gathered in the design adaptation process by intensive measurements and statistical analysis.

As a result, a new connection design based on proven elements has been developed. The interactive adjustment steps lead to a compatibility of a highly reliable connection with a completely new monitoring system generated by an iterative development process.

Downhole sensors integration process

Current development includes an innovative design for the downhole P/T gauge. The sensor is embedded into the gauge wall, with all necessary electronics integrated, supplying the sensor with the necessary power, enabling recording, and transferring the readings back to the surface (Fig. 2). Sensor selection can be tailored to field requirements and operator preferences, including pressure, temperature, vibration, or acoustic monitoring.

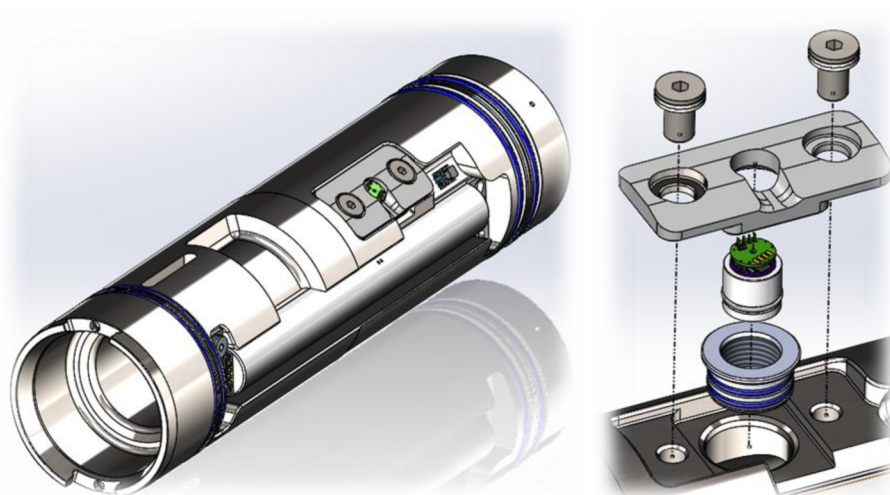


Figure 2—Integrated pressure/temperature gauge showing sensor placement within the gauge wall and embedded electronics

Surface and downhole cable connection

As the powerline for production (PLP) is a through-tubing system, a dedicated interface is required to safely transition electrical power and data between the internal downhole conduit and the surface system. To address this, a specialized component called the Cable Exit Sub was developed (Fig. 3).

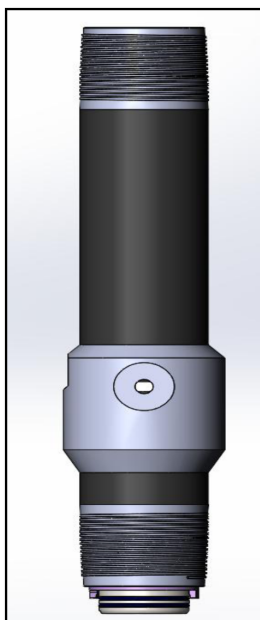


Figure 3—3D model of the Cable Exit Sub

The Cable Exit Sub is installed as part of the tubing string and is designed to:

- Provide a secure mechanical and electrical connection between the surface and downhole equipment.
- Ensure continuous electrical and data transmission from the downhole system to the surface gateway solution.
- Maintain pressure integrity using sealing elements that isolate the cable pathway from the annulus or tubing flow.

This component enables reliable surface access to downhole telemetry and power systems without compromising the mechanical or hydraulic performance of the completion.

Test results

At this stage of the project, several CFD and FEA simulations have been executed. The results of the tests are presented below:

Software simulation

Regarding CFD simulation, three different cases were examined: single-phase fluid (water reverse circulation) and multi-phase fluid with varying flow rates and phase ratios. The objective of these simulations was to determine the additional pressure drop caused by powerline for production kits.

The CFD simulation was conducted on a single tubing joint with a length of 9.75m, with pre-installed powerline for production kits. One of the simulated use cases is presented within this paper. The input data for the simulated use case are given in [Table 1](#) below.

Table 1—Input data for CFD simulation

At point X (in the well)	
Inlet pressure, MPa	20
Fluid velocity, m/s	4
Fluid viscosity, mPas	0.20083
At standard conditions	
Density of separated oil, kg/m ³	850
Gas density, kg/m ³	0.75
Water density, kg/m ³	1020
Oil flow rate, m ³ /s (m ³ /day)	2.3148×10^{-3} (200)
Gas flow rate, m ³ /s (m ³ /day)	2.3148×10^{-2} (2000)
Water cut, %	2

Prior to simulation, surface and volume meshing were executed ([Fig. 4](#) and [Fig. 5](#)).

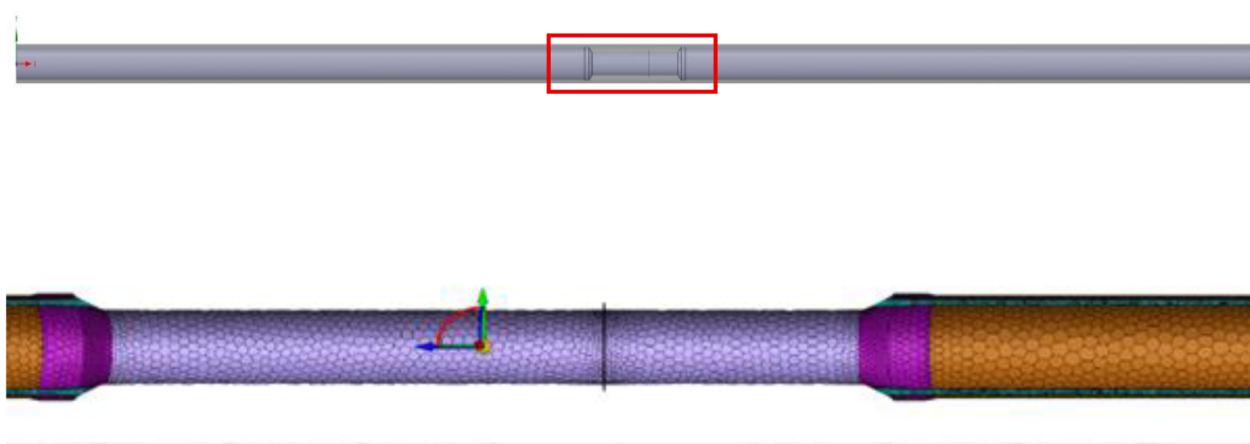


Figure 4—Hexagon Surface Mesh of the tubing joint model used in CFD simulations (227,031 faces)

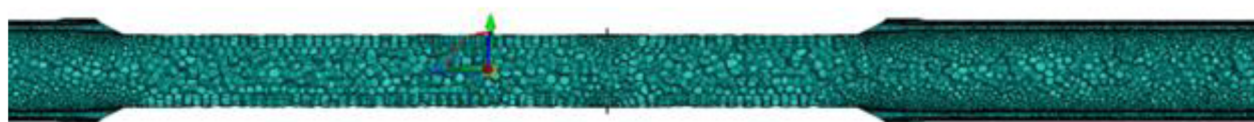


Figure 5—Polyhedral Volume Mesh of the tubing joint used in CFD simulations (1,675, 771 cells)

The results of the multi-phase fluid use case are presented in Fig. 6 and Fig. 7. As shown in the Figures below, the maximum flow velocity is reached on the section where powerline for production kits are installed, and ID is reduced. The expected pressure drops on the length of one tubing joint (9.75m) is approximately 0.0162 bar.

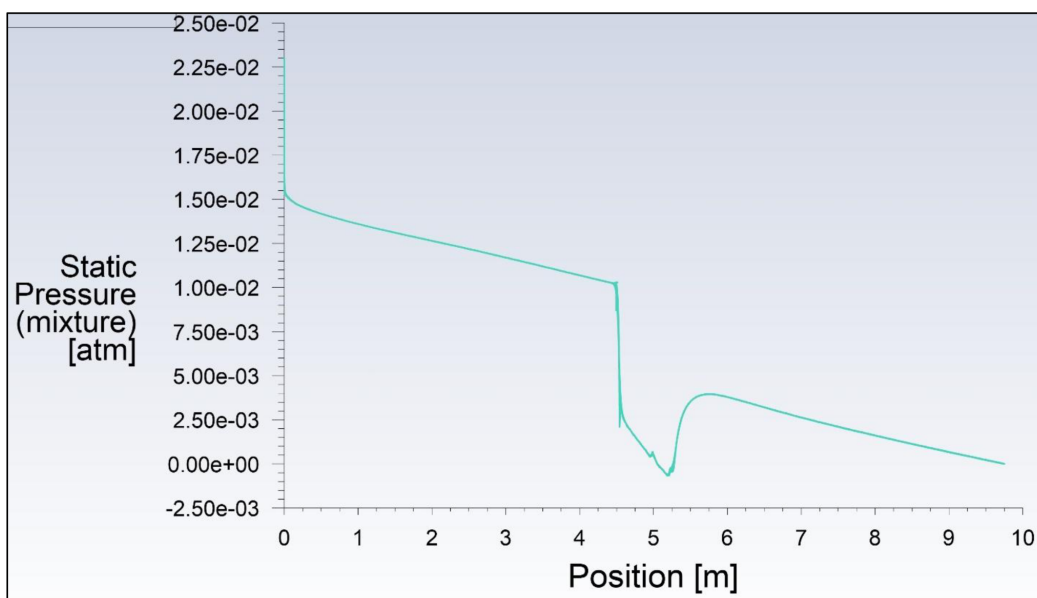


Figure 6—Simulated pressure drop profile along a 9.75m tubing joint with PLP kits installed

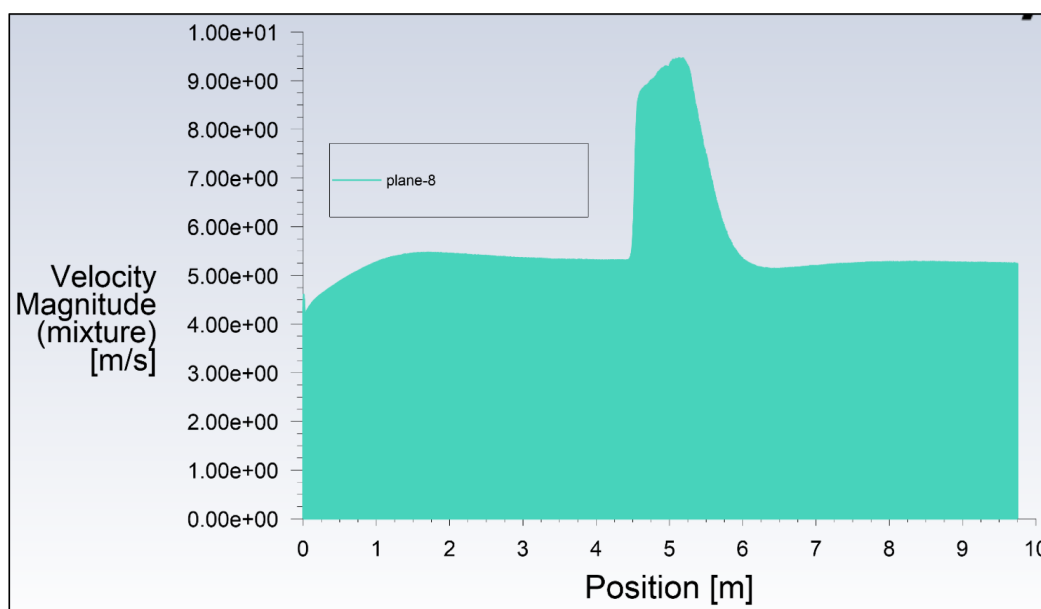


Figure 7—Multi-phase fluid velocity profile along a 9.75m tubing joint with PLP kits installed

The results of FEA simulations for the pressure/temperature gauge are presented in Fig. 8.

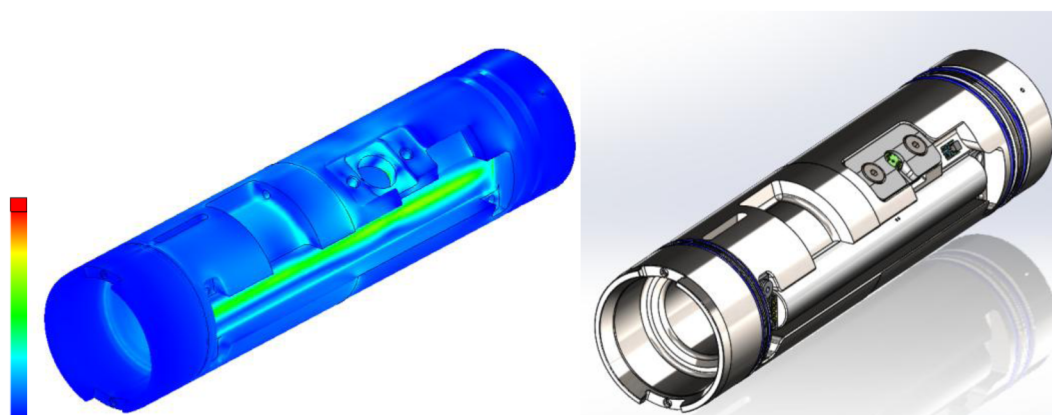


Figure 8—Finite Element Analysis (FEA) results showing stress distribution in the P/T gauge housing under simulated downhole conditions

Workshop testing

Workshop test objectives. To perform make-up and break out connection test on tubing pup joints (length: 1.9 m, OD: 5") fitted with powerline for production kits, using power tongs and a torque monitoring system. Each pair of smart pup joints will undergo at least three make-up/break-out cycles, with the lower and upper pup joints being replaced after the cycle is complete.

Testing equipment and setup. The test is to be performed using a Casing Elevator, Power Tongs, Hydraulic Power Unit Torque Monitoring System, Overhead Crane, Boom Truck, Pup joints with fitted powerline for production kits, thread compound NCS-30 ECF, and necessary electrical equipment, including a Fluke 1507 device. Powerline kits were preinstalled inside both PIN and BOX compartments of the testing pup joints.

The workshop test schematics is presented on Fig. 9.

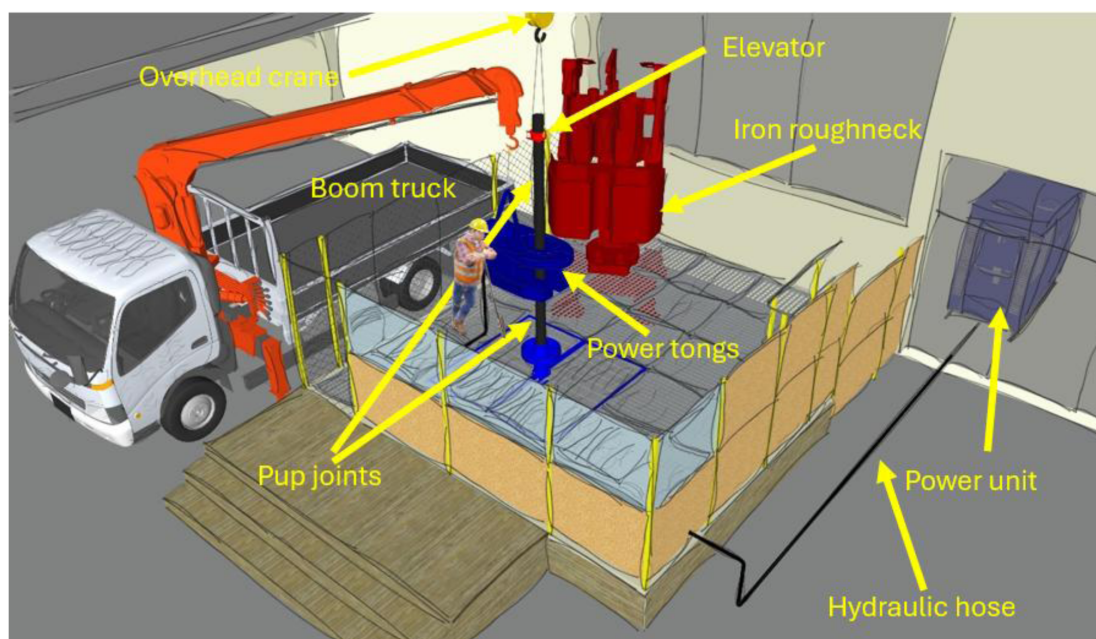


Figure 9—Schematics of planned workshop test setup

Workshop test results. Workshop test is planned as a part of the future work.

Conclusions

This paper documents the development and validation of powerline for production (PLP), an innovative through-tubing system designed to reliably deliver both electrical power (up to 300W) and high-bandwidth data (200,000 bits/sec) through production tubing.

Software simulation using ANSYS proved that the expected additional pressure drop caused by powerline for production kits is as low as 0.0162 bar/joint and does not affect the well operation.

Comprehensive pipe ID evaluation through MBG testing led to development of the new connection design.

The paper clearly shows how the PLP system can enhance smart completion reliability and installation efficiency.

Moving forward, the next step is to conduct a workshop test (TRL 5), where PLP system will be evaluated in conditions that simulate real well conditions.

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Abbreviations

API	– American Petroleum Institute
BOPD	– Barrels of Oil Per Day
CAD	– Computer-Aided Design
CFD	– Computational Fluid Dynamics
DAS	– Distributed Acoustic Sensing
DTS	– Distributed Temperature Sensing (fiber optics)
ECF	– Environmentally Compliant Fluid (in context, a type of thread compound)
ESP	– Electric Submersible Pump
FEA	– Finite Element Analysis (often interchangeably called "Finite Element Method")
HSE	– Health, Safety, and Environment
HPHT	– High Pressure, High Temperature
ICD	– Inflow Control Device
ICV	– Inflow Control Valve
IPO (valve)	– Injection Pressure Operated valve (traditional gas-lift valve type)
MBG	– Make-up/Break-out galling resistance test
MTBF	– Mean Time Between Failures
NPT	– Non-Productive Time
OPEX	– Operating Expenditure (operating costs)
PLP	– "Powerline for Production" system
POOH	– Pulling Out of Hole (pipe removal process)
P/T	– Pressure/Temperature
RIH	– Running in Hole (pipe deployment process)
TEC	– Tubing Encapsulated Cable
TRL	– Technology Readiness Level (API scale for technology maturity)

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